Supplementary Material for IAD-Risk

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1 Scope

This supplementary material documents the reproducibility boundary for the manuscript "IAD-Risk: Risk-Aware Identity-Agenda Disentanglement for Scholarly Work Deduplication." It is intended to accompany the main manuscript, not to replace the project repository or the final artifact release. The repository contains executable code, fixture data, schema contracts, and build scripts; large third-party raw data, model checkpoints, and full experimental outputs are kept outside Git and must be distributed through an artifact release when numerical result reproduction is required.

2 Reproduction Levels

Table 1: Reproduction levels and expected evidence.

Level	Required input	Evidence produced
L0	Source tree and configured Python environment	CLI help, public-release scan, unit tests, and PDF build logs.
L1	Public miniature fixtures in `tests/fixtures`	Eval-source files and an IAD-Bench fixture package with the documented schema.
L2	Independently obtained public raw files	Rebuilt DeepMatcher, SciRepEval-style, OpenAlex-style, and OpenCitations-derived evaluation sources.
L3	Released artifact package	Fixed tables, predictions, logs, manifests, checksums, and commit identifiers for result audit.

3 Environment Setup

The repository is designed to run from a standard Python environment without committing raw third-party data. Reviewers can validate the code and fixture-level reproduction path from the project root with the following commands:

```
conda create -n iad-sieve python=3.11 -y
conda activate iad-sieve
python -m pip install -e .
python -m iad_sieve.cli --help
python scripts/check_public_release.py
python manuscript/scripts/verify_fixture_rebuild.py
```

The environment check does not download full raw datasets. It verifies package installation, CLI availability, public-release boundaries, and no-network fixture rebuilding. Full numerical result reproduction still requires the L2/L3 data and artifact requirements described below.

4 No-Network Fixture Rebuild

The fixture rebuild validates the data adapters and IAD-Bench schema without depending on external network access. It is not a substitute for the full paper experiment.

```
python -m iad_sieve.cli prepare-deepmatcher \
  --table-a tests/fixtures/deepmatcher/tableA.csv \
  --table-b tests/fixtures/deepmatcher/tableB.csv \
  --pairs tests/fixtures/deepmatcher/test.csv \
  --dataset-name deepmatcher_fixture \
  --output-dir outputs/repro_fixture/deepmatcher

python -m iad_sieve.cli prepare-scirepeval-proximity \
  --metadata tests/fixtures/scirepeval/metadata.jsonl \
  --pairs tests/fixtures/scirepeval/scidocs_cite_pairs.csv \
  --dataset-name scirepeval_fixture \
  --output-dir outputs/repro_fixture/scirepeval

python -m iad_sieve.cli prepare-openalex-weak-labels \
  --works tests/fixtures/openalex/works.jsonl \
  --citations tests/fixtures/openalex/coci.csv \
  --dataset-name openalex_fixture \
  --output-dir outputs/repro_fixture/openalex \
  --min-shared-references 1 \
  --max-pairs 20

python -m iad_sieve.cli build-iad-bench \
  --source-dirs \
    outputs/repro_fixture/deepmatcher \
    outputs/repro_fixture/scirepeval \
    outputs/repro_fixture/openalex \
  --output-dir outputs/repro_fixture/iad_bench \
  --seed 42
```

The manuscript package includes a verification wrapper for the same no-network path:

```
python manuscript/scripts/verify_fixture_rebuild.py
```

The wrapper runs the four fixture commands in a temporary directory and checks that each source adapter writes `eval_documents.jsonl`, `eval_pairs.jsonl`, and `dataset_summary.jsonl`, and that the assembled IAD-Bench package writes pair, document, split, summary, provenance, and dataset-card files.

5 Public-Source Rebuild

For full data reconstruction, raw public files should be obtained from their original sources and placed under local `data/raw/` directories. These directories are intentionally excluded from Git.

The project keeps conversion commands and schema contracts under version control, while source licenses and access conditions remain governed by the original providers.

```
python -m iad_sieve.cli prepare-deepmatcher \
  --table-a data/raw/deepmatcher/DBLP-ACM/tableA.csv \
  --table-b data/raw/deepmatcher/DBLP-ACM/tableB.csv \
  --pairs data/raw/deepmatcher/DBLP-ACM/test.csv \
  --dataset-name DBLP-ACM \
  --output-dir data/processed/eval_sources/deepmatcher_dblp_acm

python -m iad_sieve.cli prepare-scirepeval-proximity \
  --metadata data/raw/scirepeval/metadata.jsonl \
  --pairs data/raw/scirepeval/scidocs_cite_pairs.csv \
  --dataset-name scidocs_cite \
  --output-dir data/processed/eval_sources/scirepeval_scidocs_cite

python -m iad_sieve.cli prepare-openalex-weak-labels \
  --works data/raw/openalex/works_cs_sample.jsonl \
  --citations data/raw/opencitations/coci.csv \
  --dataset-name openalex_cs_sample \
  --output-dir data/processed/eval_sources/openalex_cs_sample \
  --min-shared-references 1 \
  --max-pairs 5000

python -m iad_sieve.cli build-iad-bench \
  --source-dirs \
    data/processed/eval_sources/deepmatcher_dblp_acm \
    data/processed/eval_sources/scirepeval_scidocs_cite \
    data/processed/eval_sources/openalex_cs_sample \
  --output-dir data/processed/iad_bench/open_v3 \
  --train-ratio 0.8 \
  --dev-ratio 0.1 \
  --seed 42
```

6 Data Processing Traceability

Table 2 gives the audit trace from public source files to derived evaluation artifacts. The repository commits the processing commands and schema checks, but keeps the raw source files and large derived outputs outside Git. This separation allows an independent reader to rebuild the same artifact classes from obtained public sources while respecting source licenses and keeping the public repository small.

Table 2: Traceability from public data sources to derived IAD-Bench artifacts. Git contains commands, schemas, tests, and manifests; full-size inputs and outputs are verified through external artifact checksums.

Stage	Local input boundary	Processing entry point	Expected output	Git boundary
DeepMatcher gold	Raw benchmark tables and pair labels.	`prepare-deepmatcher`.	Eval documents, eval pairs, and dataset summary.	Raw tables excluded; fixtures tested.
SciRepEval proxy	Public metadata and citation-pair files.	`prepare-scirepeval-proximity`.	Proxy agenda pairs and source summaries.	Full metadata excluded; schema retained.
OpenAlex weak labels	OpenAlex works and OpenCitations COCI files.	`prepare-openalex-weak-labels`.	Silver hard negatives with provenance.	Large sources excluded; rules retained.
IAD-Bench assembly	Processed source directories from previous stages.	`build-iad-bench`.	Unified documents, pairs, splits, provenance, and dataset card.	Full package released as L2/L3 artifact.
Result audit	Prediction, score, threshold, log, and metric files.	Artifact finalization and validation scripts.	Manifest, checksums, tables, predictions, and command logs.	External artifact required.

7 Artifact Package Requirements

Full numerical result reproduction should use a released artifact package rather than raw Git-tracked files. A complete artifact package should contain a manifest, checksums, tables, predictions, configuration files, logs, and the commit identifier used to generate the results.

```
paper-artifacts/
  README.md
  manifest.json
  checksums.sha256
  configs/
  tables/
  predictions/
  reports/
  logs/
```

The manifest should record the source data version, processing commands, dependency versions, random seeds, thresholds, split configuration, hardware summary, and SHA256 checksum for every result file. The checksums should be validated before using artifact files to support claims in the manuscript.

```
python manuscript/scripts/build_artifact_release_skeleton.py \
  --output-dir /path/to/release \
  --repository-commit <commit>
python manuscript/scripts/populate_artifact_release.py \
  --artifact-dir /path/to/release \
```

```
  --source-dir /path/to/source-artifacts
python manuscript/scripts/finalize_artifact_release.py \
  --artifact-dir /path/to/release
python manuscript/scripts/validate_artifact_release.py \
  --artifact-dir /path/to/release
```

The release preparation commands create a scaffold, copy generated result files from a source artifact directory, refresh manifest and checksum metadata, and then validate the final package. The validation command checks the manifest schema, required artifact files, required result identifiers, checksum coverage, repository commit metadata, claim-boundary flags, conditional claim artifacts, and exclusion of raw third-party data, caches, credentials, local outputs, and model checkpoints.

The artifact manifest uses conditional claim flags so that optional stronger claims cannot be enabled without their evidence files. Confidence-interval claims require `bootstrap_intervals`; component-causality claims require `ablation_suite`; human-validation claims require `manual_validation_slice`; threshold-stability claims require `threshold_sensitivity_grid`; and broad method-ranking claims require the interval, validation, and threshold-grid artifacts together. If a conditional flag is set to true, the corresponding artifact row must be marked required and must include a file path and SHA256 checksum.

8 Claim-Evidence Matrix

Table 3 maps the major manuscript claims to the evidence required for responsible interpretation. The matrix is part of the submission boundary: a claim should be weakened or removed if the corresponding evidence file is absent from the released artifact package.

Table 3: Claim-evidence matrix for the main manuscript.

Claim	Required evidence	Boundary
Identity-agenda confusion is a practical false-merge risk.	Open-v2 hard-negative evaluation with FMR and HNFMR fields.	Supports a targeted scholarly deduplication failure mode, not a prevalence estimate for all domains.
IAD-Risk suppresses hard-negative false merges under the reported setting.	IAD-Risk prediction files, metric summaries, split identifiers, thresholds, and checksums.	Supports targeted false-merge suppression; does not establish broad method superiority.
IAD-Bench is provenance-aware.	Pair schema, label strength, label source, provenance fields, and fixture rebuild commands.	Supports a benchmark contract; does not turn silver or proxy labels into human gold.
Repository-level reproduction is possible without committing raw data.	L0/L1 fixture rebuild, public-release scan, unit tests, and documented public-source commands.	Verifies code and schema behavior; full numerical reproduction requires L2/L3 source files or artifacts.

9 Uncertainty and Ablation Requirements

Result packages that report confidence intervals should include bootstrap outputs generated from the exact prediction files used in the manuscript tables. The repository provides *run-iad-evidence-bootstrap* for stratified pair-level intervals and *run-bootstrap* for general metric intervals. A numerical interval should be cited only when the corresponding CSV file, command log, random seed, and checksum are present in the artifact package.

Threshold-stability claims should be tied to a released threshold-sensitivity grid generated from the same prediction files as the main tables. The grid should report threshold ranges, selected operating points, pair counts, same-work F1, FMR, HNFMR, random seeds, command logs, and checksums. Without this file, the manuscript should discuss only the fixed operating points reported in the main table.

Mechanism claims should be tied to explicit ablation outputs. The repository provides *run-iad-ablation-suite* for comparing full IAD-Risk decisions with variants that remove risk gating, agenda-non-identity blocking, or single-space separation. Without those outputs, the main manuscript should keep the mechanism claim at the level of stratified HNFMR evidence rather than claiming a complete ablation study.

10 Manual Validation Protocol

Manual validation is a future evidence layer for assessing label precision and difficult boundary cases; it is not completed in the current manuscript package. A defensible validation slice should sample 500–1,000 pairs across the strata most likely to affect claim strength: OpenAlex/OpenCitations silver hard negatives, high-score representation false-merge candidates, IAD-Risk blocked or deferred pairs, version-boundary cases, identifier conflicts, and sparse-metadata cases.

Table 4: Manual validation protocol required before stronger label-precision claims.

Protocol item	Requirement	Claim boundary
Sampling	Stratified 500–1,000 pair slice across silver hard negatives, model-disagreement pairs, version-boundary pairs, and sparse-metadata pairs.	The slice estimates difficult-case precision; it is not a prevalence estimate for all scholarly records.
Review process	two independent reviewers inspect titles, authors, venues, identifiers, abstracts, and citation context while blind to model scores and merge decisions.	Model outputs cannot determine the human label.
Label schema	Reviewers choose same work, agenda non-identity, unrelated, version boundary, or uncertain, and record a short evidence note.	Uncertain cases remain separate and are not forced into gold labels.
Adjudication	Disagreements are resolved by an adjudication log that records the final relation, rationale, and reviewer IDs or anonymous reviewer codes.	Adjudicated labels become auditable only when the log is released.
Reporting	The artifact package reports inter-annotator agreement, per-stratum counts, conflict rates, and checksums for the reviewed slice.	The manuscript must not claim human gold or stronger label precision until these files exist.

11 Claim Boundary

The main manuscript treats DeepMatcher-style labels as gold identity evidence. It treats SciRepEval and SciDocs-style proximity as same-agenda proxy evidence. It treats OpenAlex and OpenCitations-derived pairs as silver hard-negative evidence. OpenAlex and OpenCitations signals are not human gold labels. Broad method-ranking claims require stronger source-heldout evidence and manual validation beyond the current manuscript package.